**Exploratory Data Analysis & Data Cleaning Report**

**Medical Appointment No Shows Dataset**

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## 1. Dataset Overview

The *Medical Appointment No Shows* dataset contains information about patients, their medical conditions, appointment details, and whether they attended their scheduled appointment.

Each row represents **one medical appointment**, and the dataset includes both numerical and categorical features relevant to patient behavior and healthcare outcomes.

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## 2. Identification of Missing Values

Missing values were identified using the Pandas function:

```
df.isnull().sum()
```

**Observation**

Missing values were found in the following columns:

- **Age**
- **Gender**
- **AppointmentDay**
- **Scholarship**
- **Diabetes**
- **SMS_Received**

These missing values indicate incomplete patient records and inconsistencies during data collection.

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## 3. Visualization of Missing Data

Missing data patterns were visualized using a **bar chart**, showing the number of missing values per column.

**Observation**

- Numerical column **Age** had noticeable missing entries.
- Categorical columns such as **Gender** and **AppointmentDay** also showed gaps.
- No column showed extremely high missing values (greater than 50%).

This visualization helped in selecting appropriate data-cleaning strategies.

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**4. Numerical Data Imputation**

Numerical columns were handled using **median imputation**.

**Column Imputed**

- **Age**

**Reason**

- Median is preferred over mean to reduce the impact of outliers and skewed distributions.
- Ensures realistic age values without distorting the dataset.

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**5. Categorical Data Imputation**

Categorical columns were handled using **mode imputation**.

**Columns Imputed**

- Gender
- AppointmentDay
- Scholarship
- Diabetes
- SMS_Received

**Reason**

- Mode represents the most frequent category.
- Maintains consistency in categorical distributions.

- Suitable when missing values are relatively low.

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### 6. Removal of Columns with High Missing Values

Each column was evaluated for the percentage of missing values.

**Observation**

- No column exceeded the threshold of **extremely high missing values**.
- Therefore, **no columns were removed** from the dataset.

This ensured that all important features were retained for analysis.

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### 7. Dataset Validation After Cleaning

After applying all imputation techniques, the dataset was validated again.

**Observation**

- All missing values were successfully handled.
- The dataset now contains **no null values**.
- Data types and distributions are consistent.

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### 8. Before vs After Dataset Comparison

| Aspect | Before Cleaning | After Cleaning |
|---|---|---|
| Number of Rows | Same | Same |
| Number of Columns | Same | Same |
| Missing Values | Present | None |
| Data Consistency | Low | High |
| ML Readiness | ❌ Not Suitable | ✅ Suitable |

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## 9. Data Quality Assessment

### Issues Identified

- Missing values in patient demographic and appointment-related columns.
- Incomplete medical condition records.

### Issues Resolved

- Numerical missing values handled using median imputation.
- Categorical missing values handled using mode imputation.
- Dataset cleaned without loss of records.

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## 10. Final Conclusion

The *Medical Appointment No Shows* dataset initially contained missing and incomplete values that could negatively affect analysis and machine learning models. After applying systematic data cleaning techniques, the dataset became **clean, reliable, and suitable for further analysis**, including predictive modeling such as **No-Show classification**.

This cleaned dataset can now be safely used for **EDA, visualization, and machine learning tasks**.

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